



ALFRED UNIVERSITY

SCHOOL OF ENGINEERING AND APPLIED TECHNOLOGY

Department of Applied Electronics and Embedded Technology

Current Curriculum & Instruction Governance Plan

v16.3.64 operational authority over the accepted v16.3 baseline

STUDENT	William A. Green IV
ACADEMIC YEAR	2026-2027 Academic Year
DOCUMENT ID	AU-OAA-ESET301-CURR-2026-64
REVISION	3.0 / v16.3.64
STATUS	Current Governing Document

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Learn. Build. Serve. | Current runtime: v16.3.64 | Core curriculum release: 16.3

CURRENT WEBSITE USE - v16.3.64

This document is the current governance layer. The original v16.3 Curriculum & Instruction Plan remains the frozen detailed baseline for the 62 lessons. Later accepted repairs listed here supersede only the layers they explicitly changed.

1. Governing Architecture

Element	Current accepted state
Course	31 weeks / 62 primary lessons / 24 lab routes
CETa competencies	262 official competency identities preserved
Career standards	81 current traceability-reconciled standards
Semantic tasks	91 total semantic tasks; 17 Career-specific tasks in current build metadata
Integrated checks	62 integrated MCQs + 62 short responses
Gates	Weeks 5, 12, 18; Week 23 CETa readiness; Week 30 career green light
Cloud Sync	Protocol 2; local-first supported records
Runtime	v16.3.64 document-system refresh; course release remains 16.3

2. Authority Stack

Layer	Controlling record
Detailed 62-lesson baseline	AU-ESET-301-v16.3-Curriculum-and-Instruction-Plan.docx (frozen baseline)
Career wording/provenance/dependencies	v16.3.42 traceability package
Teaching Media architecture	v16.3.60 combined multimodal QA repair
Outside literature	v16.3.48 literature integration + later architecture roles
CETa Study Guide	v16.3.53 mapping through v16.3.61 built-in copy behavior
Study week selection	v16.3.62 reached-plus-current week selector
Calendar / Resume completion truth	v16.3.63 shared evidence-based stage logic
Current learner documents	v16.3.64 document-system refresh

3. 31-Week Instructional Sequence

Week	Dates	Module	Primary lessons	Lab
W01	2026-09-15 - 2026-09-19	Electricity, Safety, Ohm's Law, and Power	The four quantities and the circuit they describe / Safe bench setup, prediction, and measurement	LAB-001
W02	2026-09-21 - 2026-09-26	DC Networks, Kirchhoff's Laws, and Voltage Dividers	Series, parallel, KVL, KCL, and divider reasoning / Use expected node values to troubleshoot instead of guessing	LAB-002
W03	2026-09-28 - 2026-10-03	DMM, Bench Supply, and Oscilloscope	What each instrument measures and how the controls map to	LAB-003

Week	Dates	Module	Primary lessons	Lab
			quantities / Measurement planning, probe discipline, and uncertainty	
W04	2026-10-05 - 2026-10-10	AC Waveforms, Capacitors, Inductors, and Time Response	Waveform language and energy-storage behavior / Turn time-domain measurements into filter and fault evidence	LAB-004
W05	2026-10-12 - 2026-10-17	Foundation Mastery and Practical Fault Isolation	Interleaved foundation retrieval and error repair / A complete troubleshooting loop under a practical gate	LAB-005
W06	2026-10-19 - 2026-10-24	Reactance, Impedance, Resonance, Filters, and Transformers	From reactance to impedance, phase, and resonance / Characterize a filter or resonant network like a technician	LAB-006
W07	2026-10-26 - 2026-10-31	Components, Magnetism, Relays, Motors, and Generators	Component families, ratings, magnetism, and machines / Read a datasheet and design a relay/inductive-load interface	LAB-007
W08	2026-11-02 - 2026-11-07	Diodes, BJTs, MOSFETs, and Controlled Switching	Junction behavior and semiconductor device roles / Design and diagnose a protected low-side switch	LAB-008
W09	2026-11-09 - 2026-11-14	Rectifiers, Filters, Regulators, and Power-Supply Faults	Power-supply blocks and their expected signals / Rail-by-rail bring-up and supply troubleshooting	LAB-009
W10	2026-11-16 - 2026-11-21	Amplifiers, Op-Amps, Gain, Decibels, and Loading	Gain, dB, feedback, and ideal op-amp rules / Measure and troubleshoot a signal chain	LAB-010
W11	2026-11-23 - 2026-11-28	Number Systems, Boolean Logic, and Digital Circuits	Binary, hexadecimal, gates, and sequential state / Real logic levels, pull resistors, debounce, and measurement	LAB-011
W12	2026-11-30 - 2026-12-05	Soldering, ESD, Rework, and Version-Controlled Repair	ESD control and solder/rework fundamentals / Make physical changes traceable with Git and lab records	LAB-012
W13	2026-12-07 - 2026-12-12	Schematics, Datasheets, Signal Flow, and Root Cause	Read diagram types and turn symbols into behavior / Troubleshoot with competing hypotheses and high-information tests	LAB-013
W14	2026-12-14 - 2026-12-19	C Fundamentals and Computer Architecture	Computer blocks, representation, memory, and the build path / Write and debug small C programs methodically	LAB-014
W15	2026-12-21 - 2026-12-26	Embedded C, Bit Operations, State Machines, and Timing	Bit fields, ports, processors, and stored state / Responsive embedded architecture, volatile, and timing choices	LAB-015
W16	2026-12-28 - 2027-01-02	STM32 Project Setup, GPIO, Clocks, and SWD Debug	MCU architecture, memory map, GPIO, and debug concepts / Create, flash, verify, and document a minimum STM32 bring-up	LAB-016
W17	2027-01-04 - 2027-01-09	UART Communication and	UART frames, parameters, levels,	LAB-017

Week	Dates	Module	Primary lessons	Lab
		Python Serial Control	and error causes / Build a robust Python serial client and evidence log	
W18	2027-01-11 - 2027-01-16	I ² C, SPI, Logic Analysis, Sensors, and Actuators	Interface roles, signaling, and conversion / Datasheet-to-wire integration and bus fault isolation	LAB-018
W19	2027-01-19 - 2027-01-23	Cabling, Audio/Video Signal Paths, and Optical Systems	Media, connectors, signal paths, and optical fundamentals / Terminate, inspect, continuity-test, and isolate a connection fault	LAB-019
W20	2027-01-26 - 2027-01-30	RF, Telecommunications, Spectrum, and Technician Procedures	RF, modulation, antennas, transmission, and telecom blocks / Measurement procedure, spectrum interpretation, and technical handoff	LAB-020
W21	2027-02-02 - 2027-02-06	CETa Integration I and Mixed Technician Decisions	Interleaved recall across all CETa domains / Turn broad theory into rapid technician choices	LAB-021
W22	2027-02-09 - 2027-02-13	CETa Repair II and Project Requirements	Targeted repair and readiness evidence / Write requirements and verification before building	LAB-022
W23	2027-02-16 - 2027-02-20	CETa Readiness Gate and Reproducible Project Setup	Execute the final readiness gate without gaming it / Create a controlled project baseline before hardware bring-up	-
W24	2027-02-23 - 2027-02-27	Safe Hardware Bring-Up and CETa Power/Measurement Retention	Retention: safe power, rails, current, and instrument choices / Execute staged first power and establish a known-good baseline	-
W25	2027-03-02 - 2027-03-06	Sensor/Interface Integration and CETa Interface Retention	Retention: interface circuitry, conversion, and optical/isolation context / Integrate one peripheral through a controlled vertical slice	-
W26	2027-03-09 - 2027-03-13	Firmware Architecture, Diagnostics, and CETa Digital Retention	Retention: logic, registers, memory, and computer blocks / Separate drivers, application state, diagnostics, and recovery	-
W27	2027-03-16 - 2027-03-20	Python Hardware Test, Logging, and CETa Computer-Application Retention	Retention: computer applications, data, ports, and calculation integrity / Design a maintainable automated hardware test	LAB-023
W28	2027-03-23 - 2027-03-27	Fault Injection, Root Cause, and Technical Test Reporting	Retention: diagrams, test equipment, records, and procedure / Design controlled faults and write a defensible report	-
W29	2027-03-30 - 2027-04-03	Instrumentation, Rework, Harnesses, and Role-Proof Integration	Retention: workmanship, cabling, measurement, and safety / Assemble an employer-facing role-proof dossier	-
W30	2027-04-06 - 2027-04-10	Portfolio, Resume, Interview	Retention: mixed technical	LAB-024

Week	Dates	Module	Primary lessons	Lab
		Defense, and CETa Retention	explanations and exam maintenance / Build an evidence-first application and interview package	
W31	2027-04-13 - 2027-04-17	Application Sprint, Interview Feedback, and Continuing Mastery	Retention: protect certification knowledge after the course / Run applications as an evidence-and-feedback loop	-

4. Pedagogical Contract

- Teach from the ground up: physical meaning, units/prefixes, why the concept matters, worked examples, calculations, and links to earlier concepts.
- Concept-before-use: later labs, Career standards, and CETa reading may rely only on prerequisites that have already been taught or deliberately activated.
- Use credited authoritative diagrams/figures where helpful. AI-generated instructional diagrams are not the default instructional source.
- Use high-quality YouTube teachers when they materially improve understanding; clarity/visual explanation/pacing can outrank prestige after accuracy/currentness is acceptable.
- Active learning and changed-context evidence outrank passive consumption.

5. Teaching Media / Reading Architecture

Metric	Current value
Assignments	347
Classroom Required	247
Study	81
Engineering Library	19
Target sections	491
Inline placements	1,021
Canonical video resources	120
Canonical non-video resources	131
Central reading records	214
Required contradictions	0
Missing backlinks	0

Governing rule: complete instructional coverage first, remove unnecessary within-medium redundancy second. Every substantive section receives first-pass support in video and outside written literature where applicable. Required content is placed at the exact teaching section and remains visible centrally.

6. CETa Study Guide

Disposition	Current state
Required	64 printed pages / 14 weeks / approximately 221-260 minutes total
Study/Review	133 printed pages
Reference/Historical	27 printed pages
No-new-Required weeks	17
Authority	Current ETA scope/formulas/errata control conflicts
Mapping	Printed page +10 -> verified bundled PDF page

7. Assessment and Readiness

- Weekly Mastery normally targets $\geq 80\%$ and includes safety-critical checks where assigned.
- Lab/Application requires real route evidence plus required lab knowledge check $\geq 80\%$ when a lab exists.
- CETa readiness requires one current Alfred 100-question run $\geq 85\%$ and one separate current independent ETA/CETa practice assessment $\geq 85\%$.
- Week 18 embedded gate requires STM32, SWD, UART, I2C/SPI, logic analyzer, and fault-injection evidence.
- Week 30 career green light is evidence-based and does not require becoming engineer-level before applying for technician/test/hardware bridge roles.

8. UX / Progress Rules

- Resume Learn and Resume Study are separate.
- Study week selector permits reached weeks plus the current course week without moving Classroom progress.
- Question review routes to exact teaching sections and returns to the original question.
- Glossary: first occurrence emphasis, single-click definition, double-click glossary navigation; no hover dependency.
- Focus Prep appears at the start of sustained work. Quiet Mode reduces distraction without hiding instruction.
- Media consumption does not affect mastery. Prior-pass compatibility prevents restored Required resources from revoking historical passes.

9. Runtime / Version / Deployment Discipline

- Repository state, deployment state, and browser cache are separate facts.
- Every runtime update advances build-info.json, service-worker cache, Release Notes, and checksums together.
- Cloud Sync protocol changes only when the record protocol actually changes; v16.3.64 remains protocol 2.
- Historical reports stay frozen. Current documents label exactly which layer they govern.